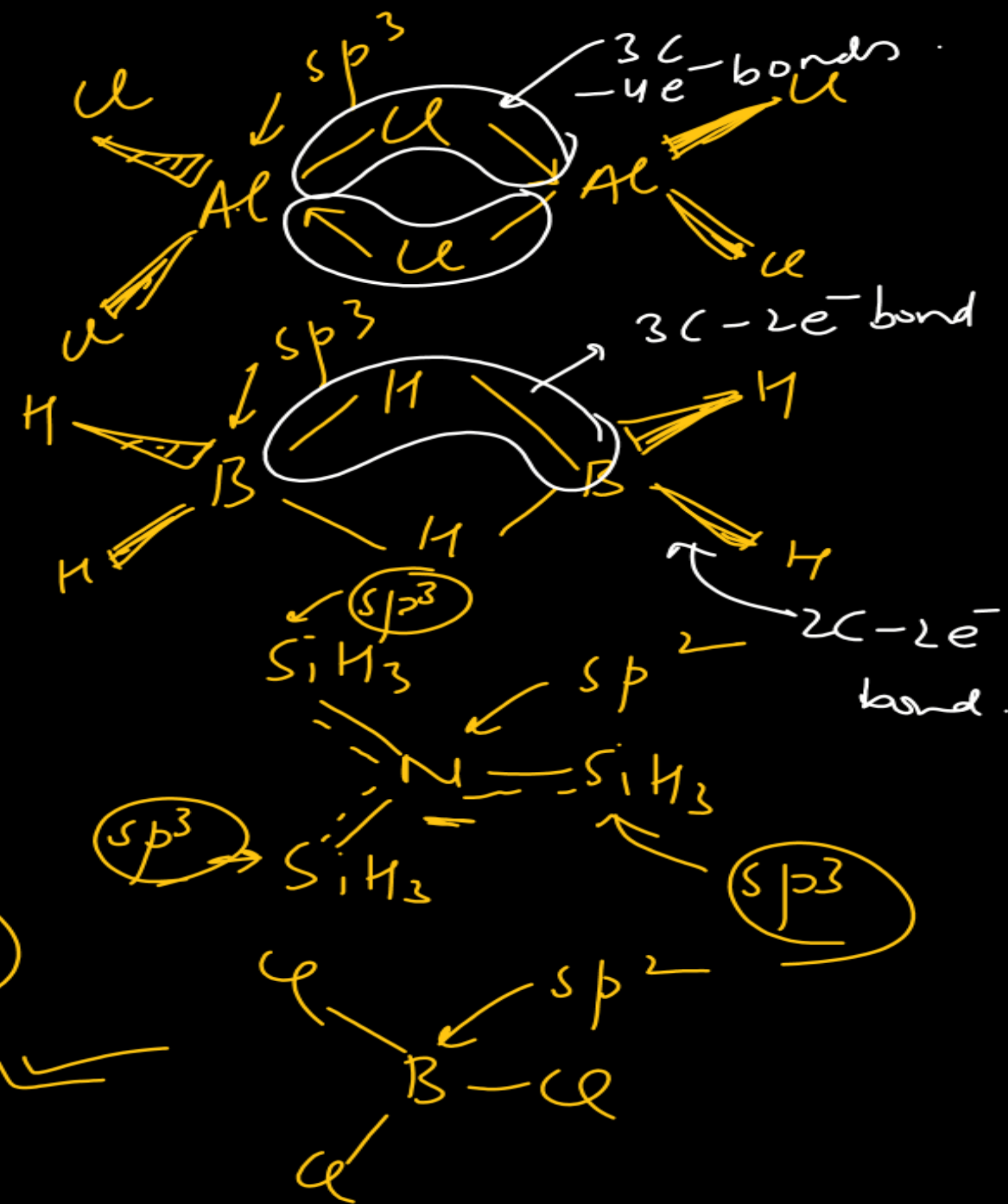


Column-1 (Molecules)	Column-2 (Geometry and Hybridisation of underline atom)	Column-3 (Specification)
(P) <u>Al</u> ₂ Cl ₆ ① (iv)	(I) Tetrahedral; sp ³	(i) 3c-2e ⁻ bond(s)
(Q) <u>B</u> ₂ H ₆ ① (i)	(II) Pyramidal; sp ³	(ii) <u>pπ - dπ back bonding</u>
(R) <u>N</u> (SiH ₃) ₃ (iii) (ii)	(III) Planar; sp ²	(iii) pπ - pπ back bonding
(S) <u>B</u> Cl ₃ (iii), (ii)	(IV) Pyramidal; sp ²	(iv) 3c - 4e ⁻ bond(s)



Select CORRECT code for species of Column-I in which all the atoms are in same plane.

- (A) (R), (II), (ii) ~~(B) (S), (III), (iii)~~ (C) (S), (III), (i) (D) (S), (IV), (iii)

Select INCORRECT code for species in which only two tetrahedron are present.

- (A) (P), (I), (iv) (B) (Q), (I), (i) ~~(C) (Q), (I), (iv)~~ ~~(D) (Q), (II), (i)~~

Select CORRECT matching option.

- (A) (R), (II), (ii) ~~(B) (R), (III), (iii)~~ ~~(C) (R), (I), (ii)~~ ~~(D) (R), (III), (ii)~~

Be₂Cl₄, Al₂Cl₆, I₂Cl₆ etc → 3c-4e⁻ bonds.

3c-2e⁻

B₂H₆, Al₂H₆, Al₂(CH₃)₆,
(BeH₂)_n

Synergic bonding in metal carbonyl complexes [M = Metal] :

(I) decreases C — O bond strength ✓

(II) increases M — C bond strength ✓

(III) increases bond order for M — C bond ✓

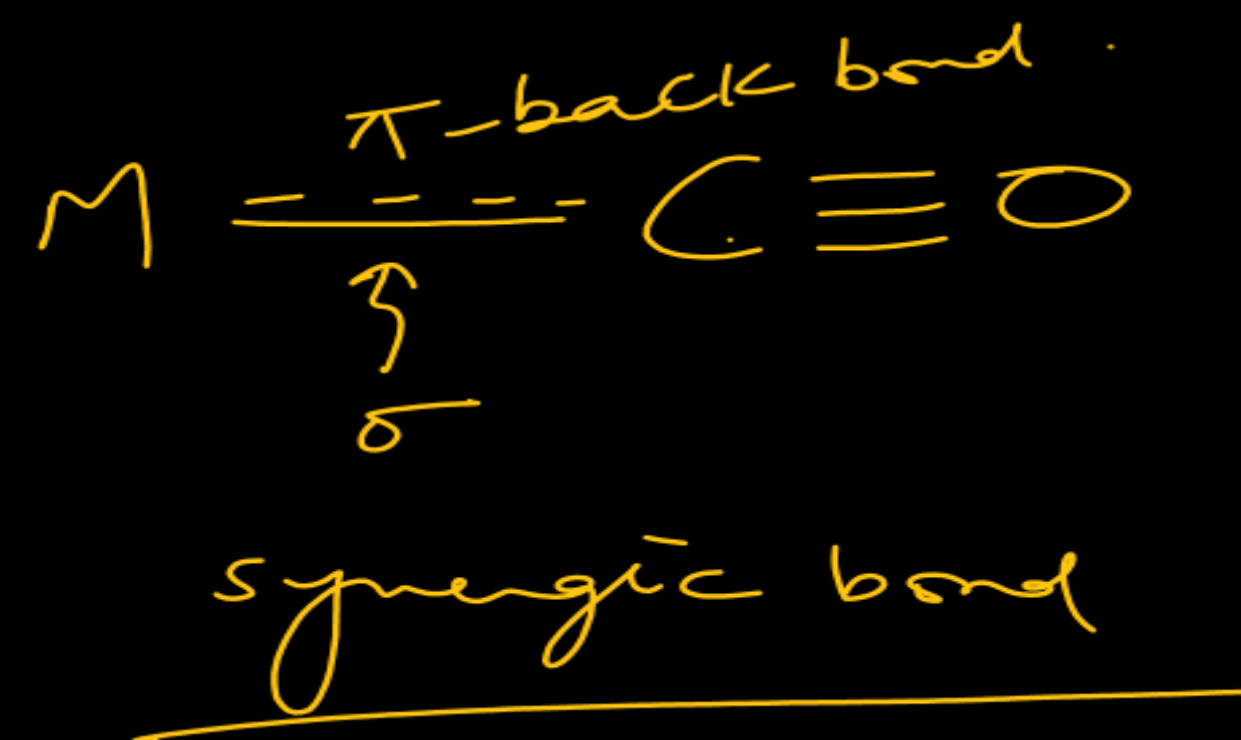
(IV) increases paramagnetism of complex ✗

(A) I, III and IV ✗

~~(B) I and III~~ ✓

(C) I and IV ✗

(D) II, III and IV ✗



Match each coordination compound in List-I with an appropriate pair of characteristics from List-II and select the correct answer using the code given below the lists.

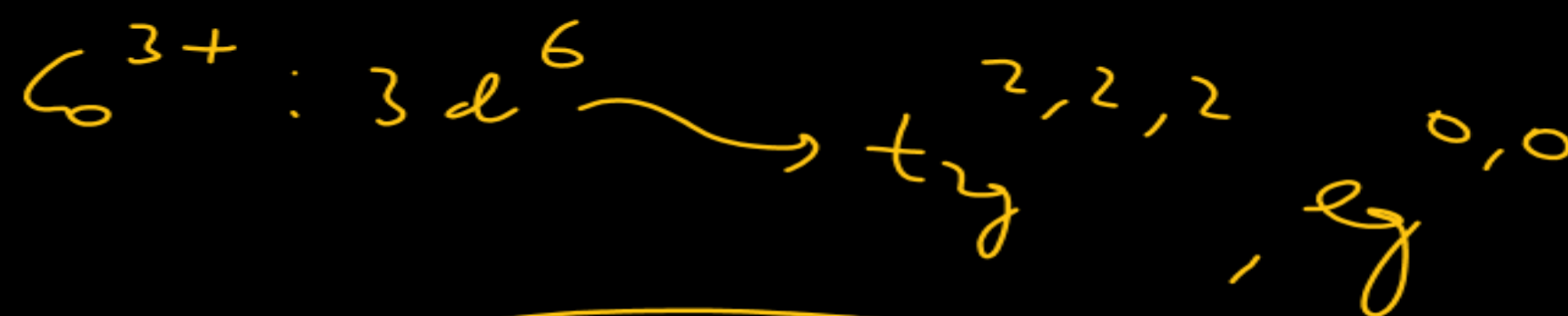
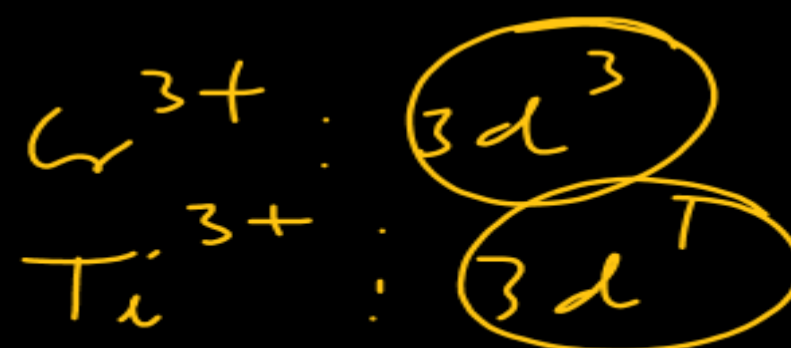
{en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$, atomic number; Ti = 22; Cr = 24; Co = 27; Pt = 78}

- | List-I | List-II |
|---|--|
| (P) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ | (1) Paramagnetic and exhibits ionisation isomerism |
| (Q) $[\text{Ti}(\text{H}_2\text{O})_5\text{Cl}](\text{NO}_3)_2$ | (2) Diamagnetic and exhibits cis-trans isomerism |
| (R) $[\text{Pt}(\text{en})(\text{NH}_3)\text{Cl}]\text{NO}_3$ | (3) Paramagnetic and exhibits cis-trans isomerism |
| (S) $[\text{Co}(\text{NH}_3)_4(\text{NO}_3)_2]\text{NO}_3$ | (4) Diamagnetic and exhibits ionisation isomerism |

Code:	P	Q	R	S
(A)	4	2	3	1
(B)	3	1	4	2
(C)	2	1	3	4
(D)	1	3	4	2

Sq. planar complex

Ans: 13



Copper is obtained from sulphide ores by roasting the ore with O_2 to form powdered copper (I) oxide and gaseous sulphur dioxide. Powdered copper (I) oxide is heated again with copper sulphide in the absence of oxygen at high temperature. Overall process is called auto reduction or self reduction and liquid metal is produced in this process.

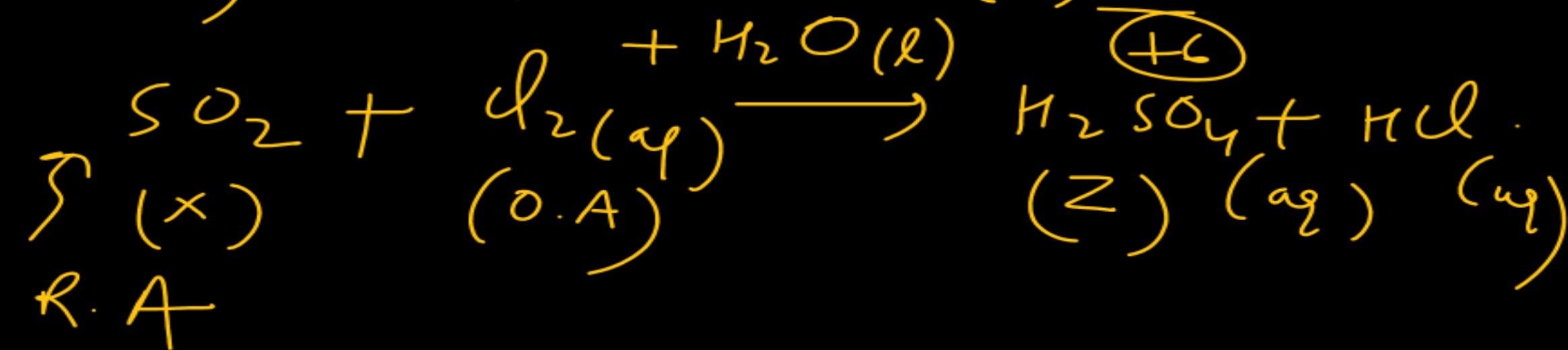
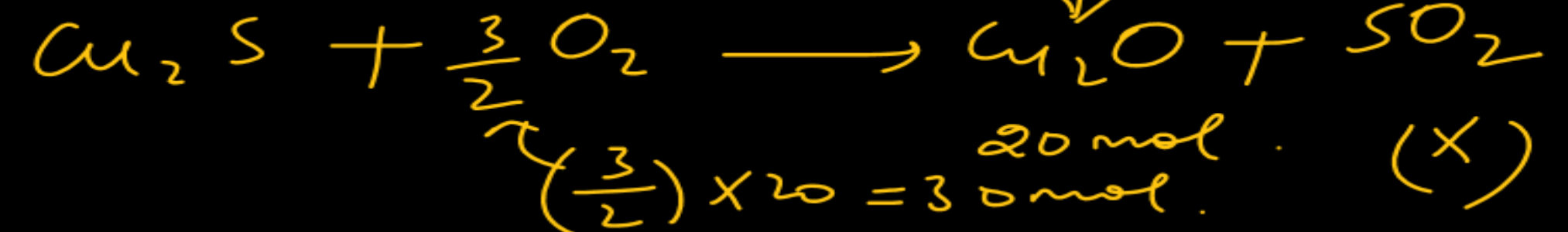
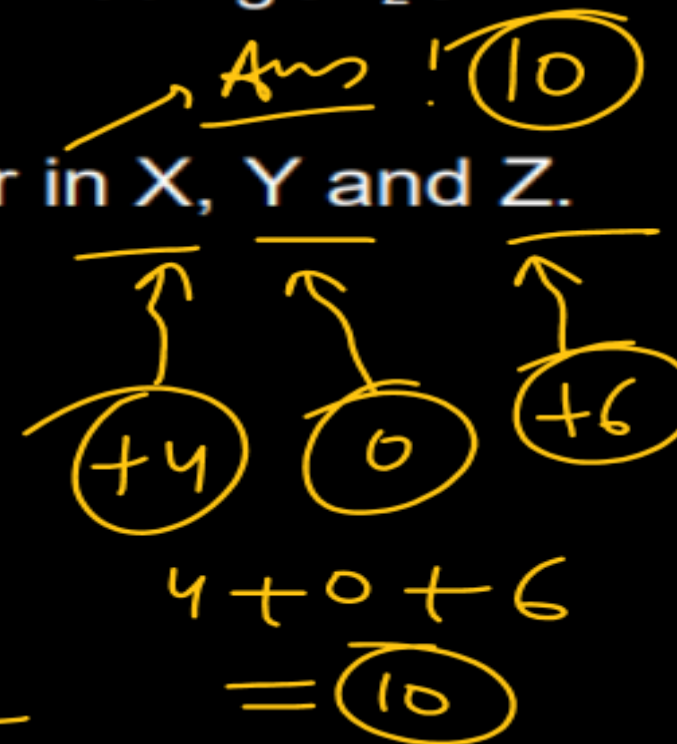
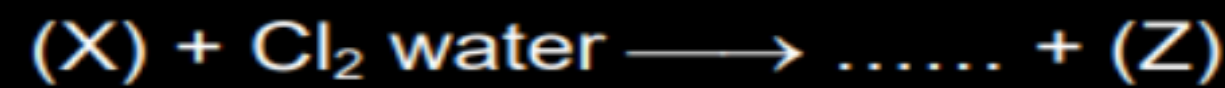
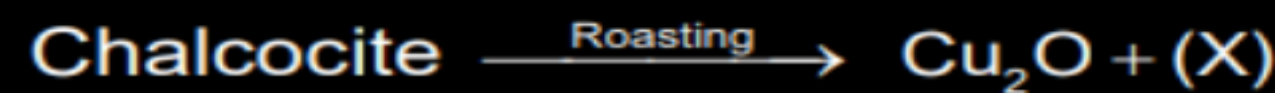
How many grams of oxygen are needed to form 2.86 kg Cu_2O from pure chalcocite ore?

$$Cu \rightarrow 63.5 \text{ g/mol}$$

$$= 20 \text{ mol}$$

$$\frac{2.86 \times 1000}{(2 \times 63.5 + 16)}$$

Find sum of the oxidation number of sulphur in X, Y and Z.



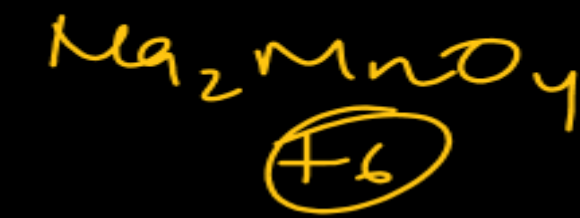
Mass of O_2 needed = 30×32

= 960 g

MnO_2 (Pyrolusite ore)

A black ore (X) on roasting with Na_2CO_3 in the presence of air gives a green compound (a).

When green compound (a) added in water it produces dark precipitate (b) and pink solution (c). Crystal of (a) on treatment with O_3 or Cl_2 completely converted to (c). Alkaline solution of (c) when treated with propene produces dark precipitate (b).



Sodium manganate

→ Geometry around central metal in (a) and (c) are respectively:

☒ (A) Tetrahedral ☒ (B) Trigonal bi pyramidal ☐ (C) Linear ☐ (D) Octahedral

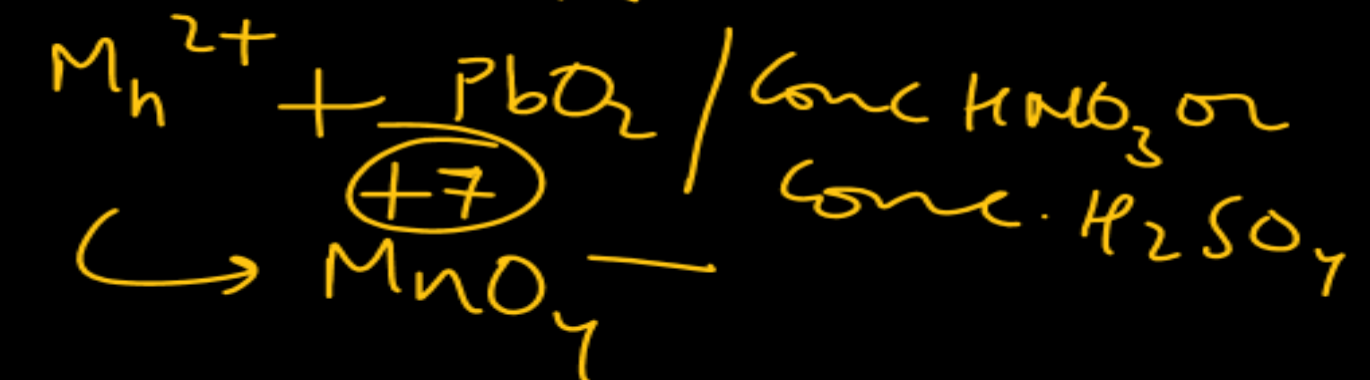
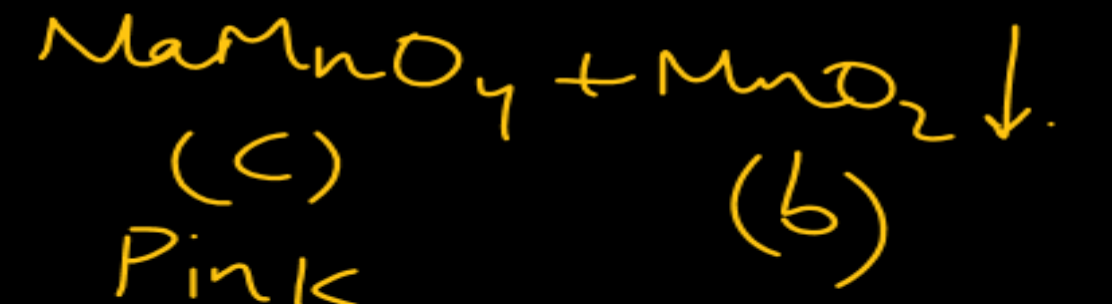
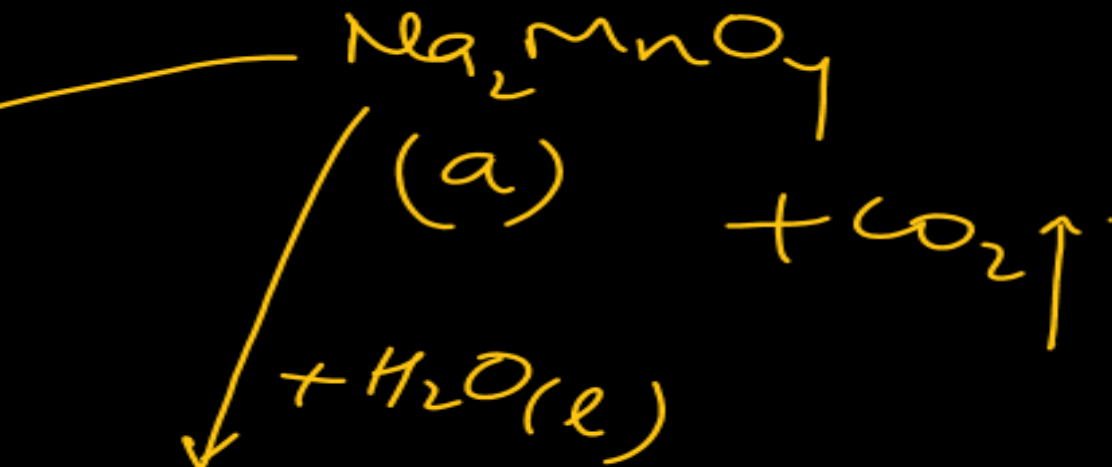
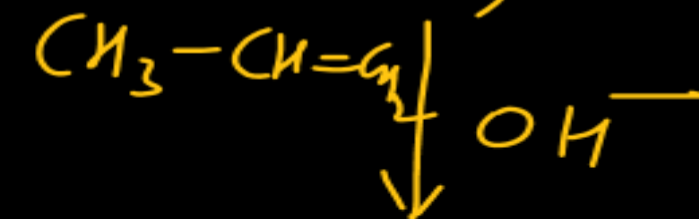
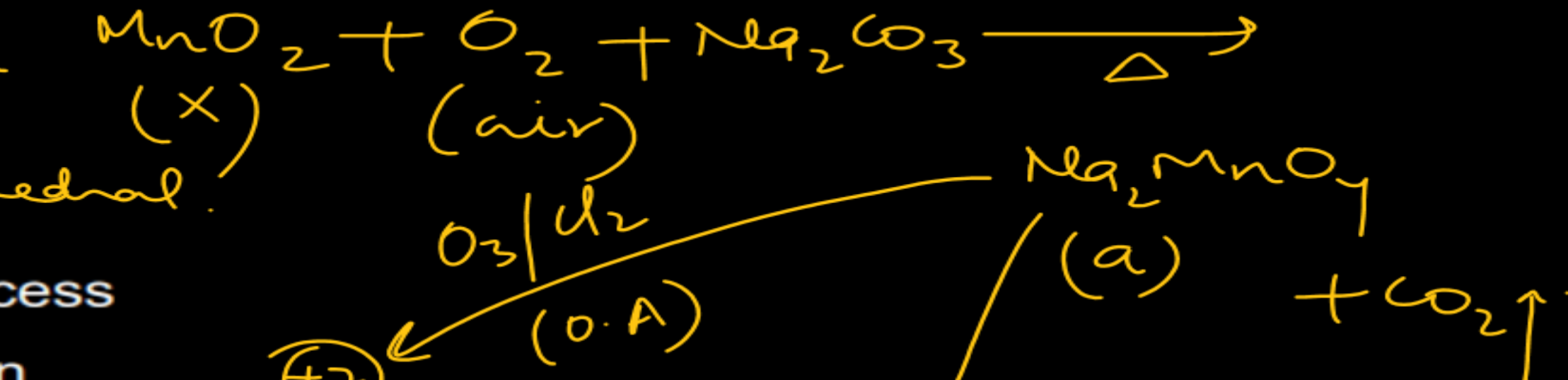
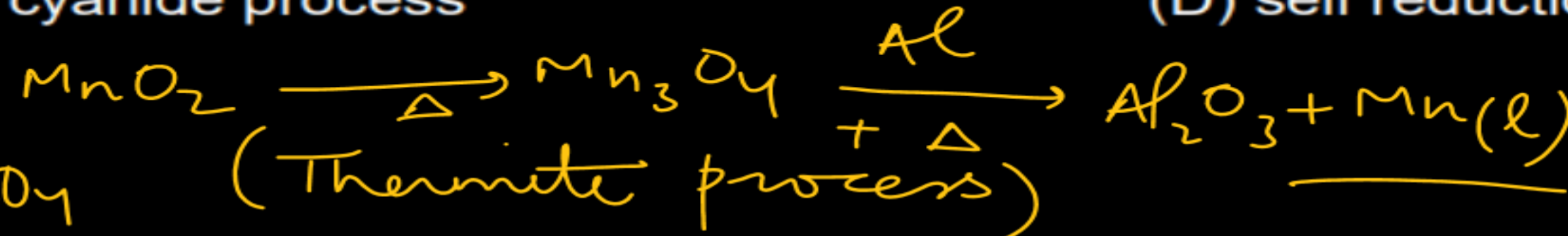
→ Metal present in ore is extracted by (X):

(A) electrolytic method

(C) cyanide process

☒ (B) thermite process

(D) self reduction



→ (c) Solution $\xrightarrow{SO_2/H^+}$ (d) Solution $\xrightarrow{Reagent(R)}$ (c) Solution

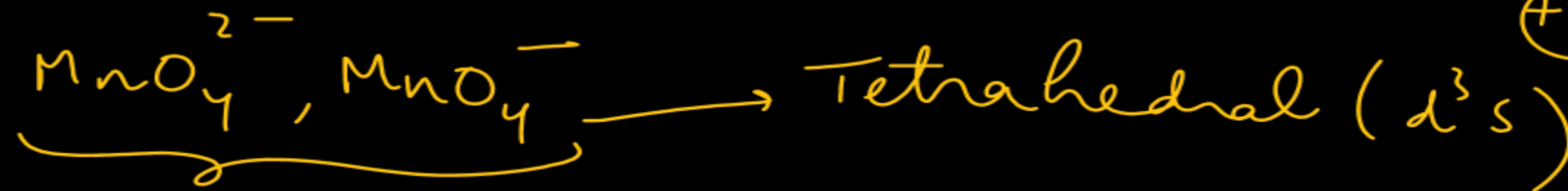
Reagent (R) can be used: $SO_4^{2-} + Mn^{2+}$ MnO_4^-

(A) OH^-/SO_3

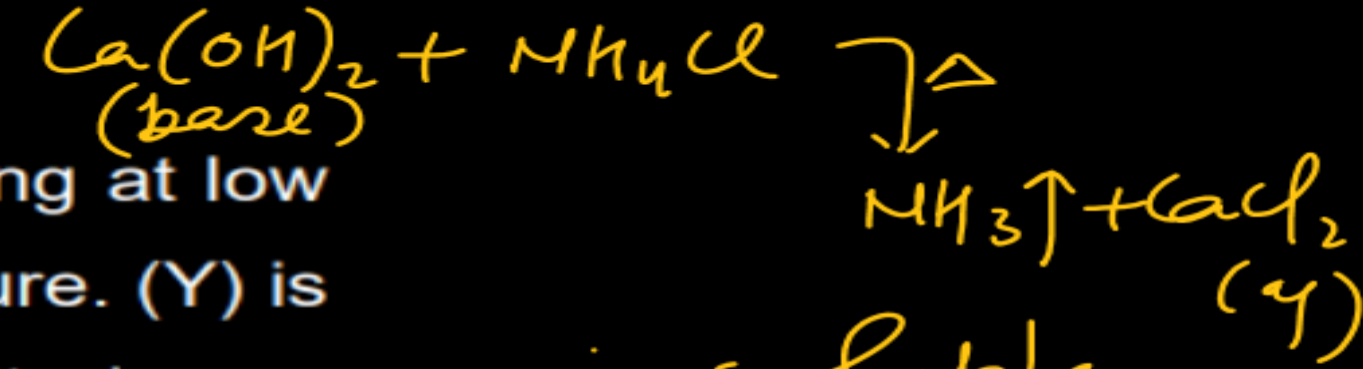
(b) H_2O_2/H^+

☒ (c) PbO_2/H^+

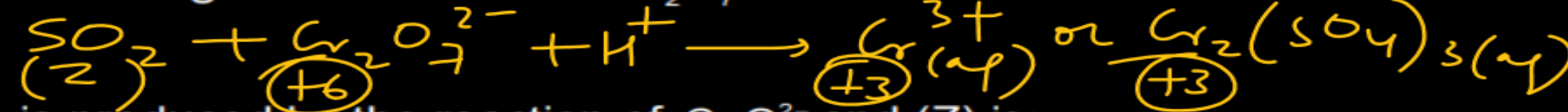
(d) Sn^{2+}/H^+



Insp ✓



A naturally occurring inorganic salt (X) produces a binary compound (Y), on heating at low temperature. A gas (Z) and a solid (S) is produced when it is heated at very high temperature. (Y) is neutral towards litmus and liquid at room temperature. Gas (Z) produces white turbidity in aq. Solution of (S) and produces green colour with $\text{Cr}_2\text{O}_7^{2-}$ solution in acidic medium.



Green compound which is produced by the reaction of $\text{Cr}_2\text{O}_7^{2-}$ and (Z) is:

- (A) Cr_2O_3 ✗ (B) K_2CrO_4 ✗ (C) K_2CrO_2 ✗ (D) $[\text{Cr}(\text{H}_2\text{O})_6]_2(\text{SO}_4)_3 + \text{H}_2\text{O(l)}$ ✓

Inorganic salt (X) is :

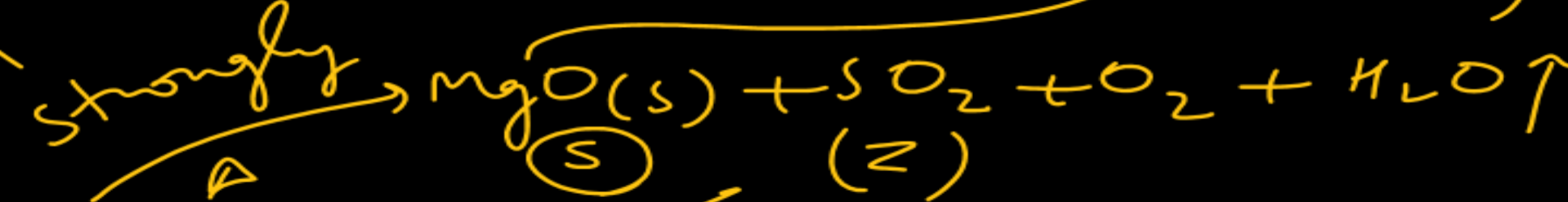
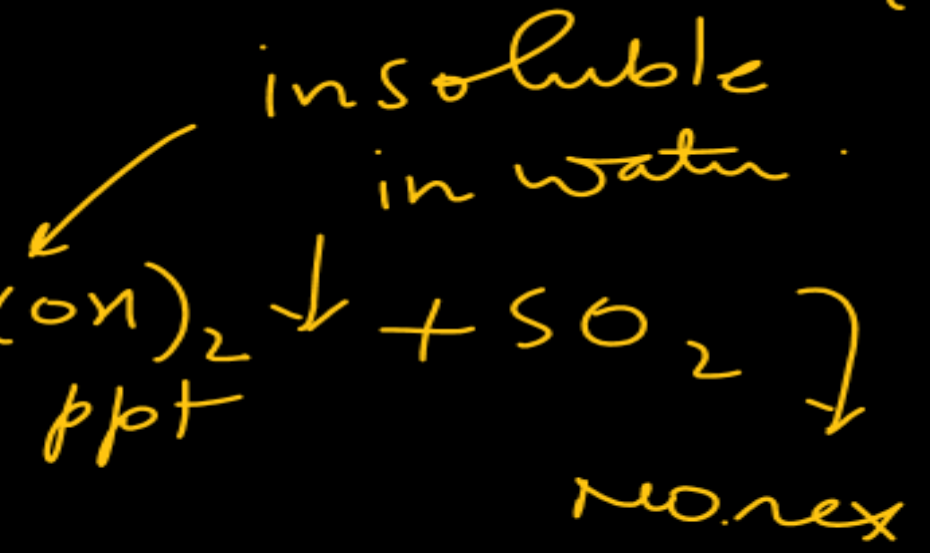
- (A) CaCO_3 ✗ (B) $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ ✗ (C) $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ ✗ (D) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ ✓

Inorganic salt (X) is :

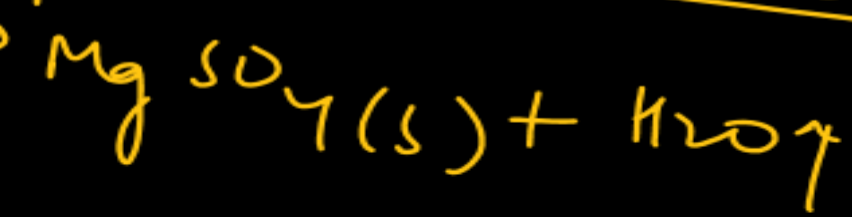
- (A) freely soluble in water ✗ (B) insoluble in water ✗ (C) sparingly soluble in water ✓ (D) soluble in hot water ✗

Aq. Solution of (S) when heated with NH_4Cl , produces:

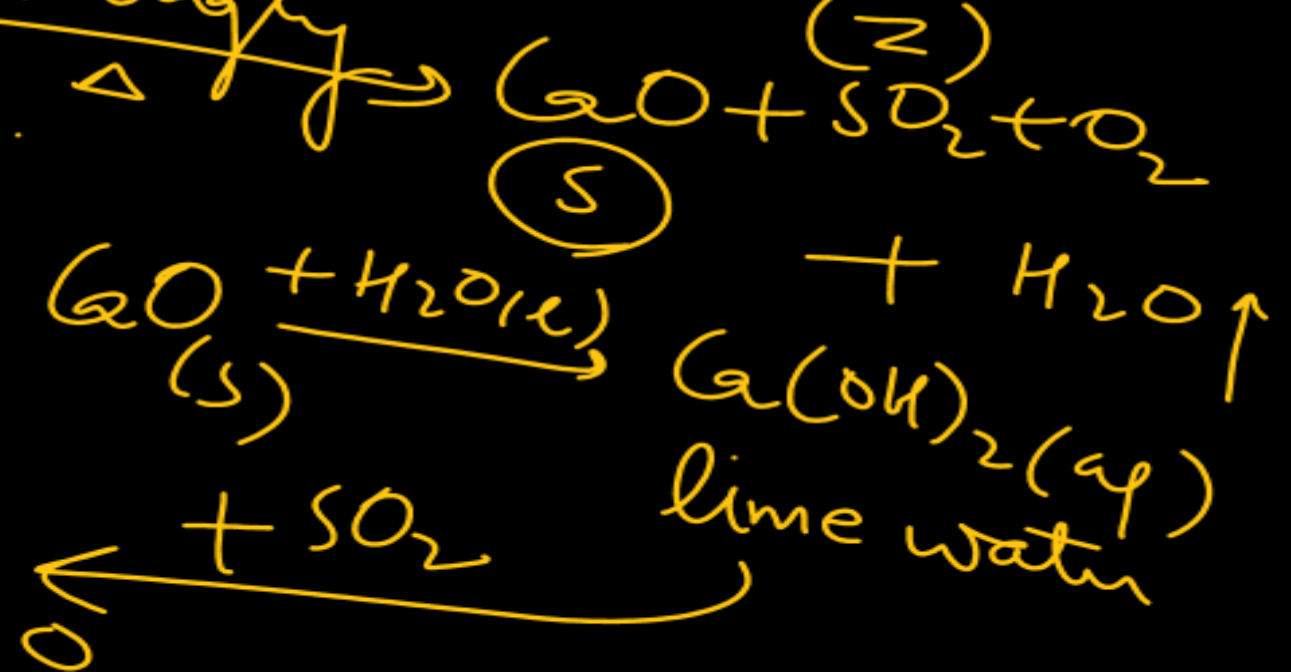
- (A) blue solution ✗ (B) white precipitate ✗ (C) ammonia gas ✓ (D) HCl gas ✗



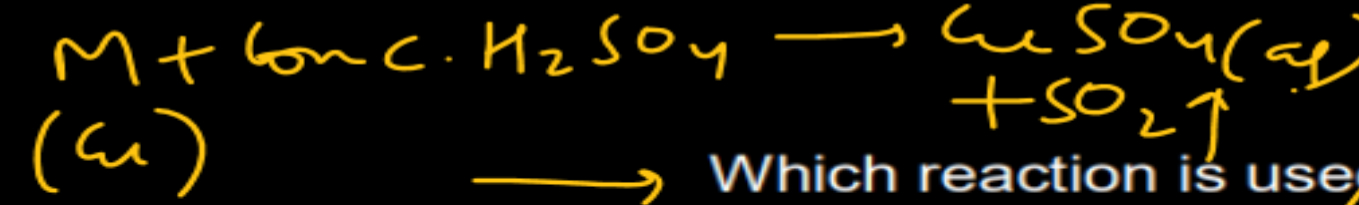
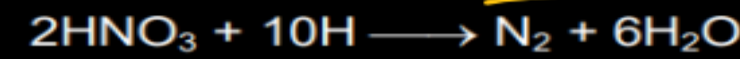
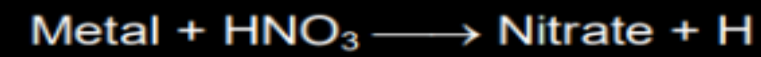
low temp



strongly



Action of nitric acid on Metals : Nitric acid reacts with most of the metals (except noble metals like gold and platinum) and non-metals. Towards its reaction with metals, HNO_3 acts as an acid as well as an oxidising agent. Like other acids, HNO_3 liberate nascent H from metals which further reduces the nitric acid into number of products like NO, NO_2 , N_2O , N_2 , NH_2OH or NH_3 according to the following reactions:



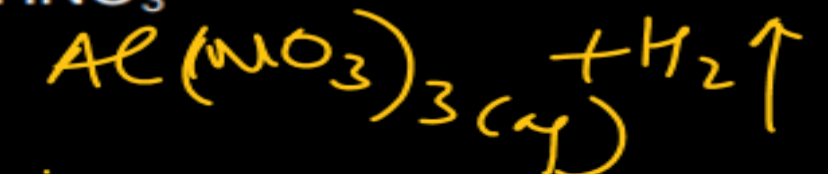
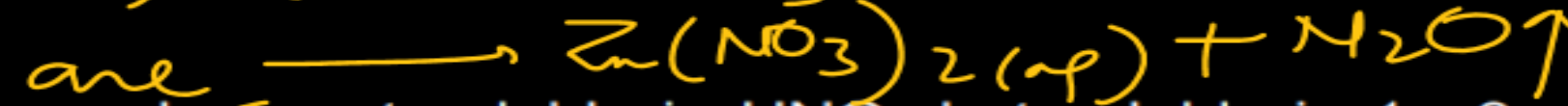
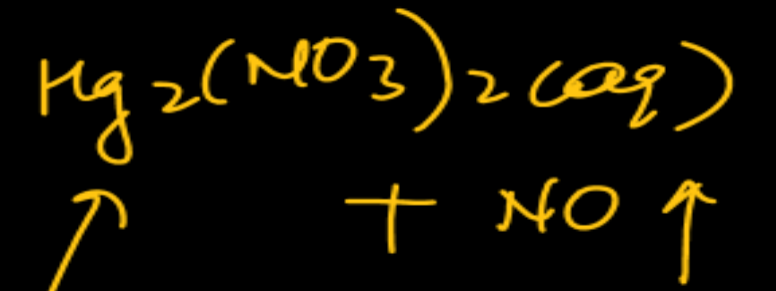
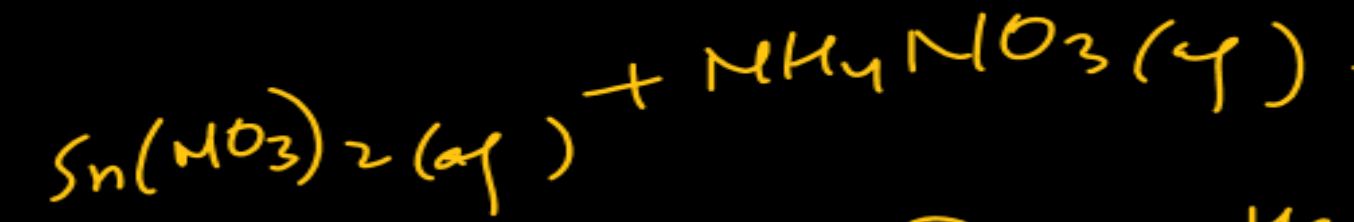
Which reaction is used to prepare laughing gas?

X (A) Reaction of Sn with very dil. HNO_3

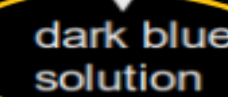
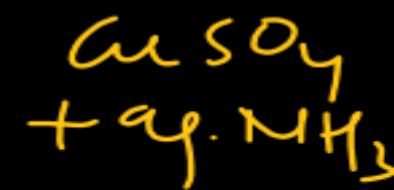
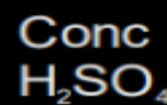
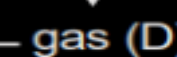
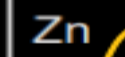
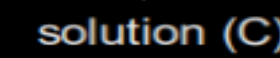
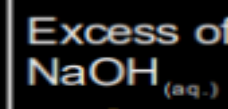
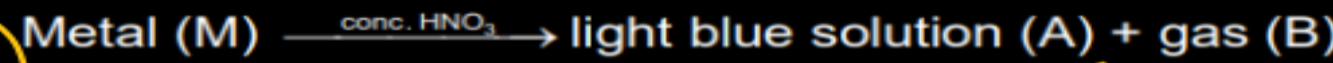
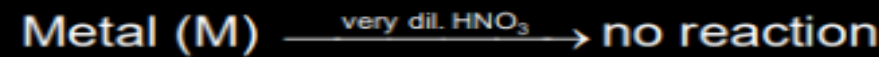
X (B) Reaction of Hg with dil. HNO_3

✓ (C) Reaction of Zn with dil. HNO_3

X (D) Reaction of Al with dil. HNO_3



Consider the following reaction sequence :



Select CORRECT statement.

(A) Gas (B) is diamagnetic X

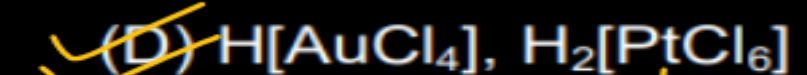
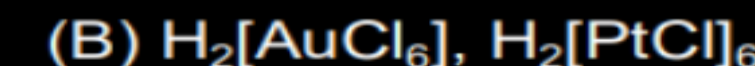
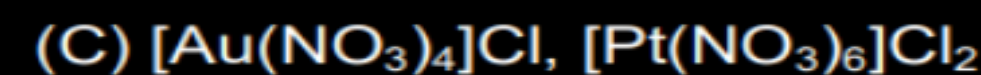
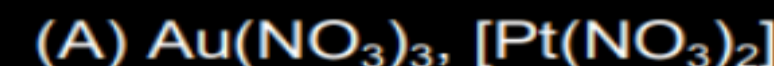
✓ (C) Dark blue solution is paramagnetic

X (B) Solution (C) contains only NaNO_2

X (D) Metal (M) is extracted by thermite process

Gold and Platinum ~~does~~ not soluble in HNO_3 but soluble in 1 : 3 mixture of HNO_3 and HCl

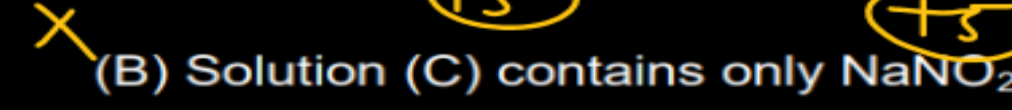
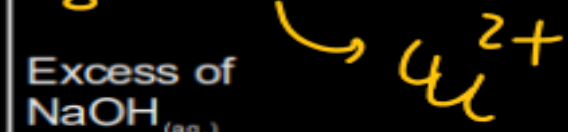
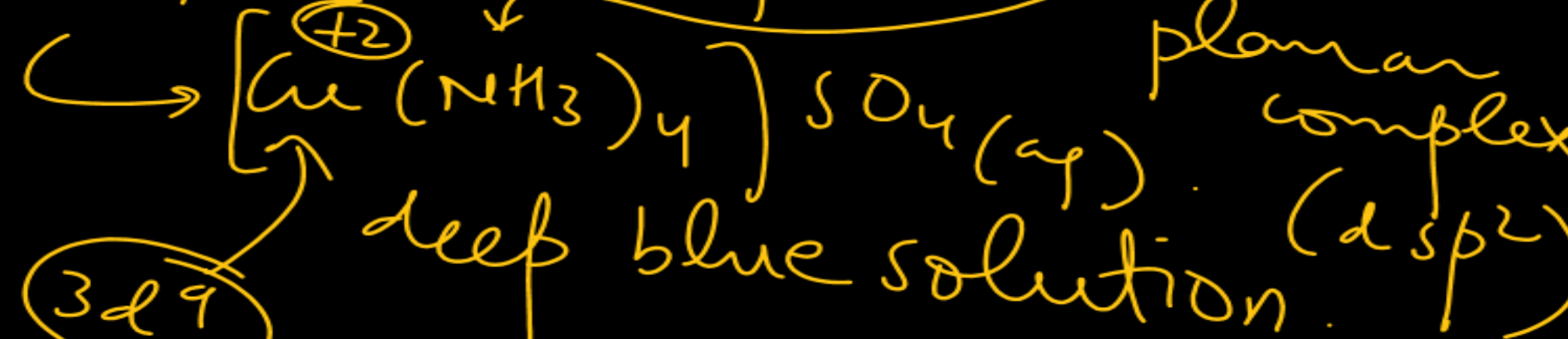
due to the formation of respectively:



aqua regia

hexachloro-platinic acid

square planar complex (dsp^2)



$\text{HNO}_3 + 3\text{HCl}$ aqua regia.
1:3 molar mixture

